

Replication and Extension of Sectoral Diversification in the UAE

HHI, DEA CCR, and an Ecuador Extension

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1 Abstract

This report replicates and extends the empirical strategy of Siddiqui and Afzal’s study of sectoral diversification in the United Arab Emirates. The replication has two components. First, it recomputes the Herfindahl-Hirschman Index (HHI) for gross fixed capital formation by sector and compares the results with the values published in the original paper. Second, it audits the paper’s DEA CCR efficiency exercise and shows that the reported universal efficiency result is consistent with an underpowered DEA design rather than clear evidence of uniformly efficient knowledge-economy pillars.

The extension applies the same diversification logic to Ecuador using value added by ISIC branch. A key methodological correction is to avoid overly aggregated three-sector HHI measures, because they impose a theoretical floor of 1/3 and mechanically classify many economies as highly concentrated. With seven mutually exclusive branches, Ecuador appears moderately diversified over 2000-2024.

2 Research Design

The original article combines descriptive knowledge-economy indicators, an HHI measure of sectoral diversification, and DEA CCR efficiency analysis. This replication keeps those three layers but separates the tasks more explicitly:

1. HHI is treated as the main reproducible diversification measure.
2. DEA is treated as a diagnostic exercise whose validity depends on the number of decision-making units (DMUs) relative to inputs and outputs.
3. The Ecuador extension tests whether the same HHI logic remains informative when the unit of analysis changes from UAE fixed capital formation to Ecuadorian value added by branch.

3 Data

For the UAE HHI replication, the project uses gross fixed capital formation by 18 sectors for 2010-2017. The published HHI values from Table 3 are stored separately from the recomputed values so that the replication comparison is transparent.

For the UAE DEA exercise, the project uses World Development Indicators for 2000-2020. The DEA dataset is built as a year-level panel where each year is a DMU. Variables are explicitly named as `input_*` and `output_*`, which keeps the DEA specification inspectable.

For Ecuador, the extension uses the UNSD National Accounts Main Aggregates dataset, specifically value added by ISIC branches. The branch set is mutually exclusive:

- Agriculture, forestry and fishing
- Mining and utilities
- Manufacturing
- Construction
- Trade, restaurants and hotels
- Transport and communication
- Other activities

4 Methodology

4.1 HHI

The HHI is computed by year as:

$$HHI_t = \sum_i \left(\frac{X_{it}}{X_t} \right)^2$$

where (X_{it}) is the value for sector or branch (i) in year (t), and (X_t) is the total across all sectors or branches in that year.

The classification follows the thresholds used in the source paper:

- $HHI < 0.15$: diversified
- $0.15 \leq HHI < 0.25$: moderately diversified

- HHI ≥ 0.25 : least diversified

4.2 DEA CCR

The project implements an input-oriented CCR DEA model. For each DMU (o), the optimization problem minimizes (θ), subject to:

$$X\lambda \leq \theta x_o, \quad Y\lambda \geq y_o, \quad \lambda \geq 0$$

An efficiency score of 1 indicates that the DMU lies on the frontier. However, DEA frontier results are fragile when there are too few DMUs relative to model dimensions. This replication uses the adequacy rule:

$$n_{DMU} \geq 3(m + s)$$

where (m) is the number of inputs and (s) is the number of outputs. Models below that threshold are classified as under-specified.

5 UAE HHI Replication

	year	hhi_computed	hhi_paper	abs_diff	classification_computed	classification_paper
0	2010	0.2364	0.2367	0.0003	Moderately diversified	Moderately diversified
1	2011	0.2366	0.2364	0.0002	Moderately diversified	Moderately diversified
2	2012	0.2479	0.2477	0.0002	Moderately diversified	Moderately diversified
3	2013	0.2432	0.2429	0.0003	Moderately diversified	Moderately diversified
4	2014	0.2431	0.2428	0.0003	Moderately diversified	Moderately diversified
5	2015	0.2492	0.2489	0.0003	Moderately diversified	Moderately diversified
6	2016	0.2446	0.2446	0.0000	Moderately diversified	Moderately diversified
7	2017	0.2468	0.2445	0.0023	Moderately diversified	Moderately diversified

The recomputed UAE HHI closely matches the published Table 3 values. In seven of eight years, the absolute difference is no greater than 0.00033. The largest difference occurs in 2017, where the recomputed value is 0.2468 and the published value is 0.2445. This difference does not alter the classification: every year remains in the moderately diversified range.

	metric	value
0	Maximum absolute difference	0.00230
1	Mean absolute difference	0.00049

The UAE results therefore support the descriptive finding that the country was moderately diversified in terms of fixed capital formation over 2010-2017, with values close to the upper bound of the moderate-diversification interval.

6 UAE DEA Audit

	model	n_inputs	n_outputs	n_data	threshold_3x	ade- quate	verdict	n_efficient	mean_efficiency
0	ICT_Knowledge_Diffusion	11	9	True	VALIDO	1	0.4183		
1	Innovation_2_to_GDP	11	9	True	VALIDO	2	0.9410		
2	Diversification_Efficiency	21	12	True	VALIDO	1	0.5181		
3	Table2_Macro_Degenerate	6	21	False	SUBE- SPECI- FI- CADO	6	1.0000		

The DEA audit is the main methodological contribution of the replication. The design closest to the original paper uses three inputs and four outputs but has only six complete annual DMUs. Its adequacy threshold is 21 DMUs. The model is therefore heavily under-specified.

	factor	efficiency_ccr_input	status
0	2015	1.0	Optimization terminated successfully. (HiGHS S...
1	2016	1.0	Optimization terminated successfully. (HiGHS S...
2	2017	1.0	Optimization terminated successfully. (HiGHS S...
3	2018	1.0	Optimization terminated successfully. (HiGHS S...
4	2019	1.0	Optimization terminated successfully. (HiGHS S...
5	2020	1.0	Optimization terminated successfully. (HiGHS S...

The under-specified model classifies all six years as efficient. This reproduces the paper's substantive pattern of universal efficiency, but the result is better interpreted as a lack of discriminatory power than as robust evidence that all pillars operate efficiently.

The redesigned DEA exercises satisfy the $(3(m+s))$ rule:

- ICT knowledge diffusion: 1 input, 2 outputs, 11 DMUs, threshold 9.
- Innovation to GDP: 2 inputs, 1 output, 11 DMUs, threshold 9.

- Diversification efficiency: 1 input, 3 outputs, 21 DMUs, threshold 12.

These valid models no longer produce universal efficiency. The number of efficient years ranges from one to two, and the average efficiency ranges from 0.4183 to 0.9410.

7 Ecuador Extension

The Ecuador extension applies HHI to value added by branch over 2000-2024.

	year	hhi	classification
0	2000	0.1824	Moderately diversified
1	2001	0.1853	Moderately diversified
2	2002	0.1932	Moderately diversified
3	2003	0.2008	Moderately diversified
4	2004	0.2010	Moderately diversified
5	2005	0.1962	Moderately diversified
6	2006	0.1949	Moderately diversified
7	2007	0.1978	Moderately diversified
8	2008	0.1962	Moderately diversified
9	2009	0.2023	Moderately diversified
10	2010	0.2000	Moderately diversified
11	2011	0.1956	Moderately diversified
12	2012	0.1991	Moderately diversified
13	2013	0.1950	Moderately diversified
14	2014	0.1956	Moderately diversified
15	2015	0.2136	Moderately diversified
16	2016	0.2168	Moderately diversified
17	2017	0.2147	Moderately diversified
18	2018	0.2171	Moderately diversified
19	2019	0.2211	Moderately diversified
20	2020	0.2367	Moderately diversified
21	2021	0.2174	Moderately diversified
22	2022	0.2124	Moderately diversified
23	2023	0.2170	Moderately diversified
24	2024	0.2135	Moderately diversified

Using seven ISIC branches is important. A three-sector measure has a theoretical HHI floor of $(1/3 = 0.333)$, which already exceeds the **least diversified** threshold of 0.25. That design mechanically removes the possibility of classifying an economy as diversified or moderately diversified. With seven branches, the floor falls to $(1/7)$, restoring discriminatory power.

	period	minimum_hhi	maximum_hhi	latest_hhi
0	2000-2024	0.1824	0.2367	0.2135

The corrected Ecuador measure remains in the moderately diversified range throughout the period. The highest value is 0.2367 in 2020, consistent with a shock year, but the series does not cross the 0.25 threshold.

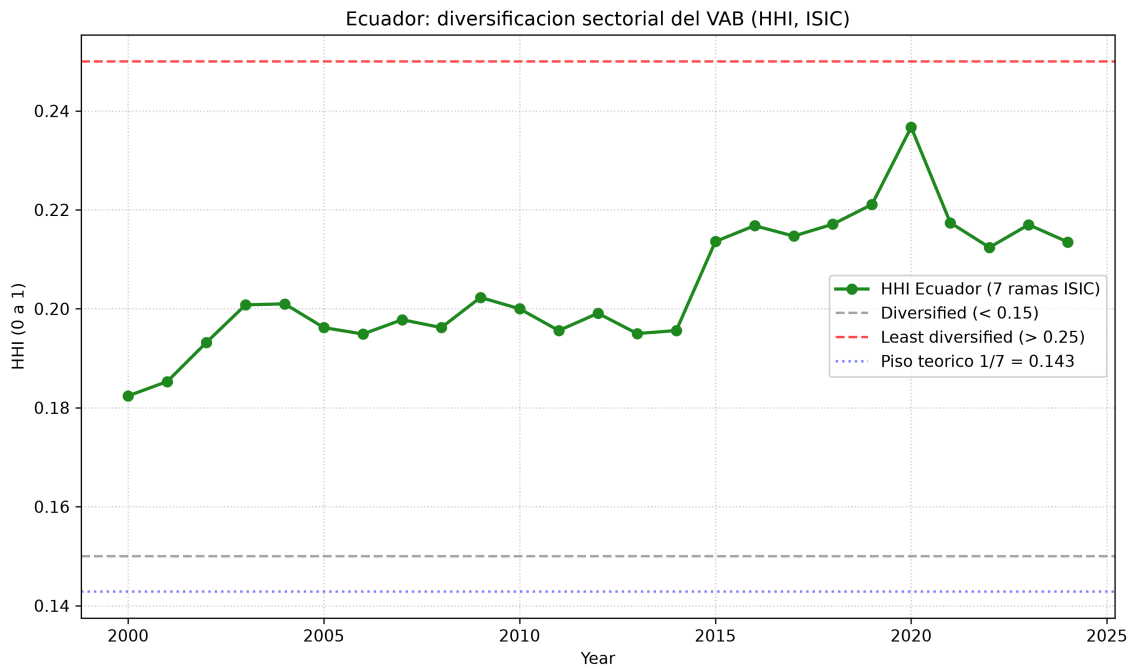


Figure 1: Ecuador HHI trend

8 Discussion

The HHI replication is successful: differences from the published UAE table are small and classification-preserving. The DEA replication is more critical. It shows that a model can reproduce the paper's all-efficient DEA result while still being methodologically weak, because too many inputs and outputs are being evaluated against too few DMUs.

The Ecuador extension adds a second methodological warning. Diversification measures are sensitive to the chosen sectoral aggregation. Too much aggregation can predetermine the classification before the data are analyzed.

9 Limitations

The UAE HHI comparison depends on the availability and versioning of sectoral fixed-capital-formation data. Minor differences may reflect source revisions, rounding, or metadata changes.

The DEA results should be read as exploratory. Year-level DMUs are useful for replication, but they are not a substitute for richer panel designs by sector, province, firm, institution, or comparable country group.

The Ecuador extension currently covers HHI only. A full DEA extension would require a separate design decision about DMUs and a defensible mapping between knowledge-economy inputs and outputs.

10 Next Steps

The most valuable next step is to build the Ecuador DEA extension. A defensible version should choose one DMU structure before collecting variables:

1. Years as DMUs, using WDI or national time-series indicators.
2. Provinces as DMUs, using Ecuadorian administrative data.
3. Sectors as DMUs, using branch-level production, employment, innovation, or export indicators.

The province or sector designs are preferable if the data are available, because they would create a richer cross-section and reduce the risk of the same low-DMU problem identified in the UAE paper.